

EXTRACT from the Technical Standard

This Extract does not replace the technical standard itself; it is only informative material about the standard.

Electronic Fee Collection (EFC) – Information exchange between service provision and toll charging

EN ISO 12855

Published in 2022 (Edition 3), 153 pages

Introduction

This technical standard (hereinafter also referred to as the “described document”) specifies the interface for the exchange of data messages between the main entities (roles) of the electronic fee collection system architecture, i.e., the toll charger and the toll service provider. It establishes the complete specification of data messages, their syntax, semantics, and the mechanism of their transmission.

Note: This Extract presents selected chapters of the described document and retains the original chapter numbering.

Usage

The described document is intended for toll chargers and toll service providers, as it establishes the basic elements of mutual interoperability at the level of their back-end systems.

Scope

The described document defines the exchange of information between the toll charger and the toll service provider. It describes the interface functionality, as well as the complete syntax and semantics of the data messages that can be exchanged via the interface (especially trusted objects, context data, exception lists, toll declarations, billing details, enforcement data). Furthermore, transmission mechanisms and support functions are described here.

Related Documents (Selection)

The described document refers to 25 technical standards, the most important of which are:

ISO 14906, Electronic Fee Collection (EFC) – Application interface definition for dedicated short-range communication (DSRC)

ISO 17573-1, Electronic Fee Collection – System architecture for vehicle-related tolling – Part 1: Reference model

ISO 17573-2, Electronic Fee Collection – System architecture for vehicle-related tolling – Part 2: Terminology

4 Abbreviations

This clause contains 42 abbreviations related to the described document, the most important of which are the following:

ADU	application data unit
DSRC	dedicated short-range communications
EFC	electronic fee collection system; electronic fee collection
GNSS	global navigation satellite system
OBE	on-board equipment
RSE	roadside equipment

TC toll charger

TSP toll service provider

Other terms and abbreviations from the ITS domain can be found in the *ITS Terminology* dictionary (www.itsterminology.org), the *StandardLand* website (www.standardland.cz) or the *OBP platform* (www.iso.org/obp).

5 Architecture

This clause, spanning 9 pages, contains a basic description of the interface functionalities for the exchange of data messages between the toll charger and the toll service provider. The following functionalities are described:

- exchange of trusted objects;
- provision of context data;
- exception list management (e.g., blacklist, whitelist);
- provision of toll declarations;
- provision of billing details;
- exchange of enforcement data;
- exchange of service quality assurance data;
- provision of medium provider billing details.

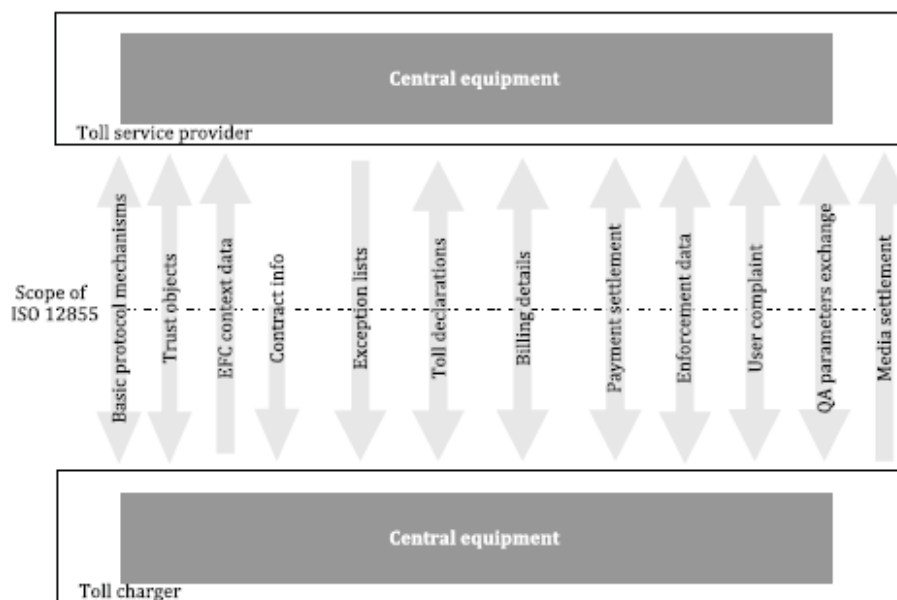


Figure 1 – Overview of functionalities (Fig. 3 of the source standard)

6 Specification

This clause, spanning 121 pages, contains a description of the structure of 19 application data units (ADUs) that form the data messages transmitted via the interface. This is the pivotal clause of the described document. The ADUs listed in the table below are defined sequentially:

Table 1 – Overview of ADUs (Tab. 5 of the source standard)

Functionality	ADU type name	Description
Basic	RequestAdu	Generic request
	AckAdu	Generic acknowledge
	StatusAdu	Generic status
Exchange trust objects	TrustObjectAdu	Send trust objects
Provide EFC context data	EfcContextDataAdu	Send EFC context data
Provide contract issuer information	ContractIssuerListAdu	Provide information on contract issuer
Manage exception lists	ExceptionListAdu	Send exception list
	ReportAbnormalObeAdu	Report abnormal OBE
Report toll declarations	TollDeclarationAdu	Toll declaration
Report billing details	BillingDetailsAdu	Billing details
Payment settlement	PaymentClaimAdu	Payment claim
	PaymentAnnouncementAdu	Payment announcement
Exchange enforcement data	ProvideUserDetailsAdu	Provide user details
	ReportCccEventAdu	Report CCC event
	ProvideUserIdListAdu	Provide a user list
Exchange QA parameters	ReportQaAdu	Report QA
Process user complaint	UserComplaintAdu	Send a user complaint
	UserComplaintResponseAdu	Reply to a user complaint
Provide media settlement data	MediaSettlementDataAdu	Send media settlement data

For illustration, the definition of the ExceptionListADU data unit is provided below.

Table 2 – Definition of ExceptionListADU (Tab. 77 of the source standard)

Field name	Data type/data description	m/o
aduIdentifier	AduIdentifier	m
exceptionListVersion	ExceptionListVersion	m
exceptionListType	ExceptionListType	m
exceptionValidityStart	GeneralizedTime	o
exceptionValidityEnd	GeneralizedTime	o
exceptionListEntries	List (ASN.1 SEQUENCE OF) of fields of type ExceptionListEntry	m
actionCode	ActionCode	o

Individual data types are explained sequentially in the text, or their definition is provided. For illustration, the definition of the ExceptionListEntry data type is provided below.

Table 3 – Definition of ExceptionListEntry (Tab. 78 of the source standard)

Field name	Data type	m/o
userId	UserId	m
statusType	ExceptionListStatusType	o
reasonCode	List (ASN.1 SEQUENCE OF) of fields of type ExceptionListReasonType	m
entryValidityStart	GeneralizedTime	o
entryValidityEnd	GeneralizedTime	o
nominalVehicleParameters	NominalVehicleParameters	o
ordinalVehicleParameters	OrdinalVehicleParameters	o
actionRequested	ExceptionListActionType	o
efcContextMark	EfcContextMark	o
mediaProviderId	Provider	o
applicableDiscounts	List (ASN.1 SEQUENCE OF) of records (ASN.1 SEQUENCE) of the following fields: discountId, proofDocuments	o

7 Transmission Mechanisms

This clause, spanning 3 pages, establishes recommendations regarding the use of a secure communication channel, data encoding, message authentication, and message signing algorithms.

Annex A (normative) – Specification of EFC Data Types

Annex A, spanning 1 page, provides the specification of the data types used according to ASN.1. A reference is provided here to the relevant ASN files, which can be imported into other application modules.

Annex B (informative) – Enforcement Process

Annex B, spanning 5 pages, describes the enforcement process. The process is illustrated using an activity diagram capturing the steps implemented on the side of the toll charger and the toll service provider, as well as the exchanged data messages.

Annex C (informative) – Data Flow Example in a Toll Domain

Annex C, spanning 3 pages, describes an example of a data flow in a toll domain. The example is illustrated using a sequence diagram capturing the data exchange between the toll charger, on-board equipment (OBE), toll service provider, and user.

Annex D (informative) – Example of Rounding

Annex D, spanning 4 pages, illustrates using the EasyGo example how rounding of amounts from aggregated toll transactions can be approached when creating billing details.

Annex E (informative) – Example of Toll Calculation Based on Context Data

Annex E, spanning 4 pages, provides an example of toll calculation based on tariff table information, toll context information, and toll domain usage information.